

EDUCATION

Columbia University

New York, U.S.A.

Ph.D. in Astronomy, Advisor: Prof. Greg L. Bryan

2015–2021

– Thesis: *Modeling the formation, evolution, and observation of first stars*

M.Phil. in Astronomy

2018

M.A. in Astronomy

2017

Indian Institute of Science Education and Research, Pune

Pune, India

B.S.-M.S. with distinction in Physics, GPA: 9.1/10

2010–2015

– Thesis: *Simulating the distribution of cosmological neutral hydrogen over cosmic times*

RESEARCH INTERESTS

- First stars, first galaxies, cosmic dawn, reionization, 21-cm signal, alternative dark matter models, computational astrophysics, semi-analytic modeling.

EXPERIENCE

Columbia University

New York, New York

Deans Fellow

Fall 2015 - Summer 2021

The University of Toledo

Toledo, Ohio

Postdoctoral research associate

Fall 2021 - Summer 2024

Georg-August-Universität Göttingen

Göttingen, Germany

Postdoctoral researcher

Winter 2024 - Present

PUBLICATIONS

1. **Kulkarni, M.**; Visbal, E.; Bryan, G.L., *Fragmentation in Population III Galaxies Formed through Ionizing Radiation*, 2019, ApJ, 882, 178. ([arXiv:1907.11724](https://arxiv.org/abs/1907.11724)).
2. **Kulkarni, M.**; Visbal, E.; Bryan, G.L., *The critical dark matter halo mass for Population III star formation: dependence on Lyman-Werner radiation, baryon-dark matter streaming velocity, and redshift*, 2021, ApJ, 917, 40. ([arXiv:2010.04169](https://arxiv.org/abs/2010.04169)).
3. **Kulkarni, M.**; Ostriker, J.P., *What is the Halo Mass Function in a Fuzzy Dark Matter Cosmology?*, 2022, MNRAS 510, 1425. ([arXiv:2011.02116](https://arxiv.org/abs/2011.02116)).
4. Shao, H.; Villaescusa-Navarro, F.; Villanueva-Domingo, P.; Teyssier, R.; Garrison, L. H.; Gatti, M.; Inman, D.; Ni, Y.; Steinwandel, U. P.; **Kulkarni, M.**; Visbal, E.; Bryan, G. L.; Anglés-Alcázar, D.; Castro, T.; Hernández-Martínez, E.; Dolag, K., *Robust Field-level Inference of Cosmological Parameters with Dark Matter Halos*, 2023, ApJ 944, 27. ([arXiv:2209.06843](https://arxiv.org/abs/2209.06843)).
5. **Kulkarni, M.**; Visbal, E.; Bryan, G.L.; Li, X., *If Dark Matter is Fuzzy, the First Stars Form in Massive Pancakes*, 2022, ApJL, 941, 18 ([arXiv:2210.11515](https://arxiv.org/abs/2210.11515)).

6. Shao, H.; de Santi, N. S. M.; Villaescusa-Navarro, F.; Teyssier, R.; Ni, Y.; Anglés-Alcázar, D.; Genel, S.; Hernquist, L.; Steinwandel, U. P.; Castro, T.; Hernández-Martínez, E.; Dolag, K.; Lovell C. C.; Visbal, E.; Garrison, L. H.; **Kulkarni, M.**, *A universal equation to predict Ω_m from halo and galaxy catalogues*, ApJ 956, 149. ([arXiv:2302.14591](#)).
7. Feathers C. R.; **Kulkarni, M.**; Visbal E.; Hazlett R., *A Global Semi-Analytic Model of the First Stars and Galaxies Including Dark Matter Halo Merger Histories*, ApJ, 962, 62. ([arXiv:2306.07371](#)).
8. Hazlett R.; **Kulkarni, M.** Visbal E.; Wise, J. H., *Calibrating a Semi-analytic Model to a Radiative Hydrodynamical Simulation of the First Stars*, 2025. ApJ, 978, 13. ([arXiv:2403.05624](#)).
9. Feathers C. R.; **Kulkarni, M.**; Visbal E., *From dark matter minihalos to large-scale radiative feedback: a self-consistent 3D simulation of the first stars and galaxies using neural networks*, 2025. JACP, 02, 43. ([arXiv:2411.07875](#)).
10. Sullivan, J.; Haiman, Z.; **Kulkarni, M.**; Visbal E., *Can supermassive stars form in protogalaxies due to internal Lyman-Werner feedback?*, 2025, Submitted to MNRAS. ([arXiv:2501.12986](#)).
11. Ambrose, A. E.; Visbal E.; **Kulkarni, M.**; McQuinn, M., *Radiative Transfer Simulations of Ly Intensity Mapping During Cosmic Reionization Including Sources from Galaxies and the Intergalactic Medium*, 2025. Submitted to JACP. ([arXiv:2502.18654](#)).

FELLOWSHIPS AND AWARDS

- **Dean's Fellowship** at Columbia University. 2015–2021
- **Junior Research Fellowship** (JRF - NET) of Council of Scientific and Industrial Research (CSIR), Govt. of India with an All India Rank of 25. 2013
- **Innovation in Science Pursuit for Inspired Research (INSPIRE)**, Department of Science and Technology, Govt. of India. 2010–2015
- **National Talent Search Examination** (NTSE) Scholarship, National Council of Education Research and Training (NCERT), India. 2008

PRESENTATIONS

- *Hydrodynamical simulations of the first stars in the cold and fuzzy dark matter cosmologies*, **HI as a cosmological probe**, Nazareth, Israel, May 2023 (talk).
- *If dark matter is fuzzy, the first stars form in massive pancakes*, **MIT Computational Structure and Galaxy Formation group meeting**, May 2023 (seminar).
- *If dark matter is fuzzy, the first stars form in massive pancakes*, **Columbia University**, New York, May 2023 (seminar).
- *Hydrodynamical simulations of the first stars in the cold and fuzzy dark matter cosmologies*, **Fermilab**, Illinois, May 2023 (seminar).
- *If dark matter is fuzzy, the first stars form in massive pancakes*, **University of Toledo**, Ohio., February 2023 (**Colloquium**).
- *If dark matter is fuzzy, the first stars form in massive pancakes*, **Tata Institute of Fundamental Research (TIFR)**, Mumbai, India, December 2022 (seminar).
- *If dark matter is fuzzy, the first stars form in massive pancakes*, **Inter-University Center for Astronomy and Astrophysics (IUCAA)**, Pune, India, December 2022 (seminar).
- *Formation of the first stars and galaxies in a fuzzy dark matter cosmology*, **AAS 240th meeting**, July 2022 (talk).
- *Population III stars and processes that delay their formation*, **AAS 237th meeting**, January 2021 (dissertation talk).
- *Population III star formation: effects of UV radiation, baryon-dark matter streaming velocity, and redshift*, **Galaxies discussion group**, **University of Cambridge**, October 2020 (talk).

- *The critical dark matter halo mass for Population III star formation: dependence on Lyman-Werner radiation, baryon-dark matter streaming velocity, and redshift*, **The First Stars, SAZERAC**, October 2020 (talk).
- *A critical mass for Pop III stars: dependence on LW radiation, dark matter-baryon streaming and redshift*, **SAZERAC**, July 2020 (poster).
- *A critical mass for Pop III stars: dependence on LW radiation, dark matter-baryon streaming and redshift*, **First Stars VI**, Concepción, Chile, March 2020 (poster).
- *A critical mass for Pop III stars: dependence on Lyman-Werner radiation, baryon/dark-matter streaming, and redshift*, **AAS 235th meeting**, Honolulu, Hawaii, January 2020 (talk).
- *Fragmentation in Ionized Pop III Galaxies*, **Into the Starlight: The End of Cosmic Dark Ages**, Aspen, Colorado, March 2019 (talk).
- *Fragmentation in Ionized Pop III Galaxies*, **Cosmology: the Next Decade**, International Centre for Theoretical Sciences, Bengaluru, India, January 2019 (talk).
- *Fragmentation in Ionized Pop III Galaxies*, **Astrophysical Frontiers in the Next Decade and Beyond**, Portland, Oregon, June 2018 (poster).
- *Fragmentation in Ionized Pop III Galaxies*, **Enzo workshop**, Atlanta, Georgia, May 2018 (talk).
- *Commissioning of a new 15-m radio telescope at NCRA*, **Astronomical Society of India's** annual meeting, Pune, India, February 2015 (poster).

TEACHING

- **Teaching Assistant** at Georg-August-Universität Göttingen Winter 2024
Introduction to General Relativity
- **Guest lectures** at the University of Toledo Spring 2021-Fall 2021
on numerical methods in the class Computational Physics
- **Guest lectures** at the University of Toledo Fall 2021
on Planetary Geology in the class Solar System Astronomy
- **Co-Instructor** at Columbia University Fall 2017 - Spring 2020
Modern Cosmology course for the Science Honors Program for high school students
- **Teaching Assistant** at Columbia University Spring 2018
Modeling the Universe.
- **Instructor** at Columbia University Fall 2017
Astronomy Lab 1: Earth, Moon and Planets.
- **Observational Teaching Assistant** at Columbia University Spring 2017
Setting up and helping with the observing sections of Astronomy labs.
- **Instructor** at Columbia University Fall 2016
Astronomy Lab 2: Stars, Galaxies and Cosmology.

MENTORING

- **Colton Feathers** Graduate student at the University of Toledo. Fall 2021–present
- **Ryan Hazlett** Graduate student at the University of Toledo. Fall 2021–present
- **Abigail Ambrose** Graduate student at the University of Toledo. Fall 2021–present
- **James Sullivan** Graduate student at Columbia University. Fall 2022–present
- **Thomas Behling** Undergraduate student at the University of Toledo. Summer 2023–present

PUBLIC OUTREACH

- Public Outreach talk on ‘Clocks of the Universe’ September 2018
A part of the Columbia astronomy outreach lectures series. Covered in press [1](#), [2](#).
- Regular volunteer for Columbia astronomy outreach events 2015–2018
Night sky observations using telescopes.
- Regular volunteer for [Rooftop Variables](#) at Columbia University 2015–2021
A group for interacting with high school astronomy clubs in the New York City area.
- Volunteer at [Reading Team Math](#) 2017–2018
After school program for math education for young children.

SKILLS

- **Languages:** Python (13 years), Cython (7 years), C (7 years).
- **Tools:** ENZO, GADGET-2, MUSIC, YT, YTREE.

[Github profile](#)

REFERENCES

1. Prof. Greg L. Bryan
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2. Prof. Eli Visbal
Department of Physics and Astronomy,
University of Toledo,
Toledo, OH, U.S.A.
email: elijah.visbal@utoledo.edu
3. Prof. Jens Niemeyer
Institut für Astrophysik und Geophysik,
Georg-August-Universität Göttingen,
Göttingen, Germany.
email: jens.niemeyer@phys.uni-goettingen.de
4. Prof. Zoltan Haiman
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